

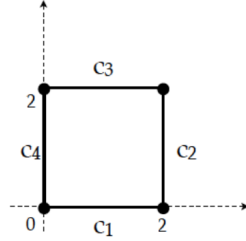
Exam *Variational methods applied to modeling.* May 2024.

Calculators and cell phones are prohibited.

Handwritten and lecture notes are allowed.

Exercise 1 (Unisolvent finite element)

Let the square $K = [0, 2] \times [0, 2]$. We denote by c_i , $i = 1, \dots, 4$, the four edges of K :



We consider the space $\mathcal{P} = \text{span}\{1, x, y, xy\}$ and the following set of linear functionals in $\mathcal{P}$:

$$\Sigma = \left\{ p \rightarrow \int_{c_1} p \, d\sigma, p \rightarrow \int_{c_2} p \, d\sigma, p \rightarrow \int_{c_3} p \, d\sigma, p \rightarrow \int_{c_4} p \, d\sigma \right\}.$$

Is the finite element $\{K, \Sigma, \mathcal{P}\}$ unisolvent? Depending on your answer, find a non zero function in $\mathcal{P}$ that vanishes for all the elements in Σ or find the basis functions of this finite element.

Exercise 2 (2D variational formulation)

Let Ω be an open bounded subset of $\mathbb{R}^2$ with a regular boundary $\partial\Omega$ such that $\partial\Omega = \Gamma_n \cup \Gamma_r$ with $\Gamma_n \cap \Gamma_r = \emptyset$. Let $a > 0$, $b > 0$, $f \in L^2(\Omega)$ and $g \in L^2(\Gamma_r)$. We consider the problem

$$(P) \begin{cases} -\Delta u + u = f, & \text{in } \Omega, \\ \partial_n u = a, & \text{on } \Gamma_n, \\ \partial_n u + bu = g, & \text{on } \Gamma_r. \end{cases}$$

1. Find a variational formulation (VF) associated to the problem (P).
2. Prove existence and uniqueness of a solution to this variational problem (VF).
3. Show that there exists a constant $C > 0$ such that

$$\|u\|_{H^1(\Omega)} \leq C \left(\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\Gamma_r)} + a \right).$$

Exercise 3 (1D variational formulation)

We consider the bilinear form defined on $H_0^1([-1, 1])$ by

$$a(u, v) = \int_{-1}^1 \left(u'(x)v'(x) + u(x)v(x) - \lambda u(0)v(x) \right) dx,$$

where $\lambda \in \mathbb{R}^*$ is fixed.

1. (a) Prove that $|u(0)| \leq \|u'\|_{L^2(]-1,1])}$ for all $u \in H_0^1(]-1,1])$.

Hint: For $u \in H^1(]-1,1])$, we remind that $u(y) = u(x) + \int_x^y u'(t)dt$ for all $(x, y) \in [-1, 1]^2$.

- (b) Deduce that if $|\lambda| < \sqrt{2}$ then the bilinear form a is coercive.

Hint: We remind that, for every pair of real numbers α and β , we can write $\alpha\beta \leq \frac{1}{2}(\alpha^2 + \beta^2)$.

2. Prove that for all $f \in L^2(]-1,1])$, if $|\lambda| < \sqrt{2}$ then it exists a unique solution $u \in H_0^1(]-1,1])$ to the problem

$$a(u, v) = \int_{-1}^1 f(x)v(x) dx, \quad \forall v \in H_0^1(]-1,1]).$$

3. Prove that if $|\lambda| < \sqrt{2}$ then the problem

$$\begin{cases} -u''(x) + u(x) - \lambda u(0) = f(x), & x \in]-1, 1[, \\ u(-1) = u(1) = 0, \end{cases}$$

admits a unique solution in $H^2(]-1,1])$.

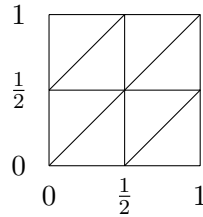
4. Let $\mu \in \mathbb{R}^*$. Following the same reasoning than in question 1, propose a sufficient condition on μ and λ such that the bilinear form defined on $H_0^1(]-1,1])$ by

$$\tilde{a}(u, v) = \int_{-1}^1 \left(e^{\mu x} u'(x) v'(x) + \cos\left(\frac{\pi}{3}x\right) u(x) v(x) - \lambda u(0) v(x) \right) dx$$

is coercive.

Exercise 4 (P1 finite element method)

Let Ω be the unit square $]0, 1[\times]0, 1[$ meshed in the following way



We consider the variational problem

$$\text{Find } u \in V \text{ such that } \int_{\Omega} \nabla u(x, y) \cdot \nabla v(x, y) dx dy = \int_{\Omega} v(x, y) dx dy, \quad \forall v \in V,$$

where $V = \{u \in H^1(\Omega), \gamma_0(u) = 0 \text{ on }]0, 1[\times \{0\}\}$. We denote by $u_h \in V_h$ the approximate solution of u obtained by a $\mathbb{P}_1$ finite element method using the simple mesh described above and by $AU = L$ the associated linear system.

1. Guess which boundary value problem is associated to this variational formulation.
2. Define the approximate space V_h and determine its dimension.
3. On a figure of the mesh, introduce a numbering of the degrees of freedom. Using this numbering, determine which entries are zeros in the matrix A .
4. We denote by ϕ_c the (global) basis function associated to the central node $(\frac{1}{2}, \frac{1}{2})$.

(a) Show that $\int_{\Omega} \phi_c(x, y) dx dy = \frac{1}{4}$.

(b) Compute $\int_{\Omega} |\nabla \phi_c(x, y)|^2 dx dy$.

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Exercise 1 (Unisolvent finite element)

Let T be the reference triangle (with vertices $a_1 = (0,0)$, $a_2 = (1,0)$ and $a_3 = (0,1)$). We consider the finite element $(T, \mathcal{P}, \Sigma)$ where $\mathcal{P} = \text{span}(1, x, y)$ and $\Sigma = \{p \rightarrow p(\frac{1}{3}, 0), p \rightarrow p(0, \frac{1}{3}), p \rightarrow \int_T p(x, y) dx dy\}$. Is this finite element unisolvent? Depending on your answer, find a non zero function in $\mathcal{P}$ that vanishes for all the elements in Σ or determine the basis functions of this finite element.

Exercise 2 (2D variational formulation)

We consider the following problem defined on the unit square $\Omega =]0, 1[^2$ and for a function $f \in L^2(\Omega)$: Find $(x, y) \rightarrow u(x, y)$ solution of

$$(P) \quad \begin{cases} -\partial_{xx}u - \partial_{yy}u - \partial_{xy}u = f, & \text{in } \Omega, \\ \partial_y u + \frac{1}{2}\partial_x u = x, & \text{on } \Gamma_1 = \{(x, 0), 0 \leq x \leq 1\}, \\ \partial_x u + \frac{1}{2}\partial_y u = y, & \text{on } \Gamma_2 = \{(1, y), 0 \leq y \leq 1\}, \\ u = 0, & \text{on } \Gamma_3 = \{(x, 1), 0 \leq x \leq 1\}, \\ u = 0, & \text{on } \Gamma_4 = \{(0, y), 0 \leq y \leq 1\}. \end{cases}$$

1. First, we consider the case of homogeneous Dirichlet boundary conditions on the entire boundary $\partial\Omega$. Write a variational formulation associated to the problem

$$(\tilde{P}) \quad \begin{cases} -\partial_{xx}u - \partial_{yy}u - \partial_{xy}u = f, & \text{in } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$$

introducing a **symmetric** bilinear form on $H_0^1(\Omega) \times H_0^1(\Omega)$.

2. Now, we are going back to the boundary conditions of (P) . What is the natural space V to describe this problem by a variational formulation?
3. Write a variational formulation associated to (P) using a symmetric bilinear form on $V \times V$. We will precise carefully the corresponding linear form on V .
4. Study the existence and the uniqueness of a solution of this variational formulation, making clear the norm on V that you use.

Exercise 3 (Convection-diffusion problem)

Let $f \in L^2(]0, 1[)$ and ν and β two positive numbers. We consider the following problem

$$(P) : \quad -\nu u''(x) + \beta u'(x) = f(x), \text{ in }]0, 1[, \quad u(0) = u(1) = 0.$$

1. Write the variational formulation (FV) associated to (P) using an appropriate Hilbert space V . We will not make an integration by parts for the second term of the bilinear form.
2. Prove that $\int_0^1 u'(x)u(x)dx = 0$ for all $u \in H_0^1(]0, 1[)$ and then show that (FV) has a unique solution.

We now consider a uniform discretization $(x_i)_{i=0,\dots,N}$ of the domain $[0, 1]$, made of N intervals of length $h = \frac{1}{N}$. Let V_h be the finite dimension subspace of V that is used for a $\mathbb{P}_1$ finite element approximation.

3. Write the approximate variational formulation (FV_h) making explicit the definition of V_h and then show that the final linear system to solve is of the form

$$AU = F$$

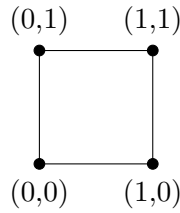
with $A = (a_{i,j}) \in \mathcal{M}_{N-1}(\mathbb{R})$ defined such that $a_{i,j} = 0$ if $|i - j| > 1$ and

$$\begin{aligned} a_{i,i-1} &= \frac{\nu}{h}(-1 - \frac{\gamma}{2}) & \text{for all } i = 2, \dots, N-1, \\ a_{i,i} &= 2\frac{\nu}{h} & \text{for all } i = 1, \dots, N-1, \\ a_{i,i+1} &= \frac{\nu}{h}(-1 + \frac{\gamma}{2}) & \text{for all } i = 1, \dots, N-2, \end{aligned}$$

where we have introduced the Péclet number $\gamma = \frac{h\beta}{\nu}$.

Exercise 4 ($\mathbb{Q}_1$ finite element method)

1. We consider the reference square $\hat{K}$ as shown in the following figure.

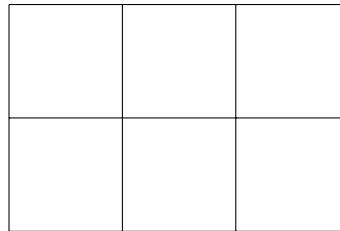


The nodes are numbered anticlockwise from $(0, 0)$. Let $\{\hat{\phi}_1, \hat{\phi}_2, \hat{\phi}_3, \hat{\phi}_4\}$ be the set of local basis functions on $\hat{K}$ using a $\mathbb{Q}_1$ finite element method (each of these functions are in $\text{span}(1, x, y, xy)$). After computations, the local stiffness matrix is given by

$$\left(\int_{\hat{K}} \nabla \hat{\phi}_i(\hat{x}, \hat{y}) \cdot \nabla \hat{\phi}_j(\hat{x}, \hat{y}) \, d\hat{x} d\hat{y} \right)_{1 \leq i, j \leq 4} = \frac{1}{6} \begin{pmatrix} 4 & -1 & -2 & -1 \\ -1 & 4 & -1 & -2 \\ -2 & -1 & 4 & -1 \\ -1 & -2 & -1 & 4 \end{pmatrix}.$$

Make the computations to recover **one** entry of this matrix (the one that you prefer).

2. Let $\Omega =]0, 3[\times]0, 2[$ meshed as shown in the following figure



We consider the problem

$$\begin{cases} -\Delta u = 1 & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}.$$

Assemble the global stiffness matrix associated to this mesh and this problem approximated by a $\mathbb{Q}_1$ finite element method.